Units, Physical quantities, and vectors

- Predicting the path of a thunderstorm helps reduce damage to life and property.
- The thunderstorm is moving with a speed of 20 km/h.
- Its direction is 53° north of east.
- The time considered is 1 hour.
- The problem asks for the distance moved toward the north in 1 hour.

Physics

- Physics is one of the most fundamental sciences.
- Ideas of physics are used by scientists in many fields such as chemistry, paleontology, and climatology.
- Physics forms the foundation of all engineering and technology.
- Engineering designs (TVs, spacecraft, tools, etc.) require understanding basic physical laws.
- Studying physics is challenging but also rewarding and intellectually satisfying.
- Physics explains everyday and natural phenomena (blue sky, radio waves, satellite motion).
- Physics represents a major achievement of human intellect in understanding nature.
- The chapter introduces essential preliminary concepts for studying physics.

➢ Topics include:

- Nature of physical theories
- Use of idealized models
- Systems of units
- Accuracy and uncertainty in measurements
- Use of rough estimates when exact answers are not practical
- Basics of vectors and vector algebra. Vectors are important for describing physical quantities with both magnitude and direction (e.g., velocity, force).

The Nature of Physics

- Physics is an experimental science.
- Physicists observe natural phenomena.
- They look for patterns that relate different phenomena.
- These patterns are known as physical theories.
- Well-established and widely accepted theories are called physical laws or principles.

Caution: The meaning of the word “theory

- A theory is not a random or unproven idea.
- It is a well-supported explanation of natural phenomena.
- Theories are based on observations and accepted fundamental principles.
- A scientific theory is developed through extensive research.
- Example: Theory of biological evolution, established through generations of observation and study.

The Nature of Physics

- Developing a physical theory involves:
- Asking appropriate questions.
- Designing experiments.
- Drawing correct conclusions from results.
- Galileo Galilei studied the motion of falling objects.
- He concluded that the acceleration of a falling body is independent of its weight.
- The development of physical theories is often indirect and includes mistakes and revisions.
- Physics is not just a set of facts, but a process of discovering general principles.

- No physical theory is considered absolutely final or perfect.
- Conflicting observations can disprove a theory, but it can never be proven completely true.
- Galileo’s theory appeared incorrect when comparing a feather and a cannonball in air.
- The difference in falling rates is due to air resistance and buoyancy, not weight.
- In a vacuum, both objects fall at the same rate, confirming Galileo’s idea.
- Every theory has a limited range of validity.
- Galileo’s theory applies when air forces are much smaller than the object’s weight.
- Later theories can extend the range of validity of earlier ones.
- Newton’s laws expanded and improved Galileo’s work on falling bodies.

Standards and units

Fundamental Quantities	SI Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	S
Electric current	Ampere	A
Temperature	Kelvin	K
Amount of subtance	mol	mol
Luminous Intensity	candela	cd

Unit Prefixes

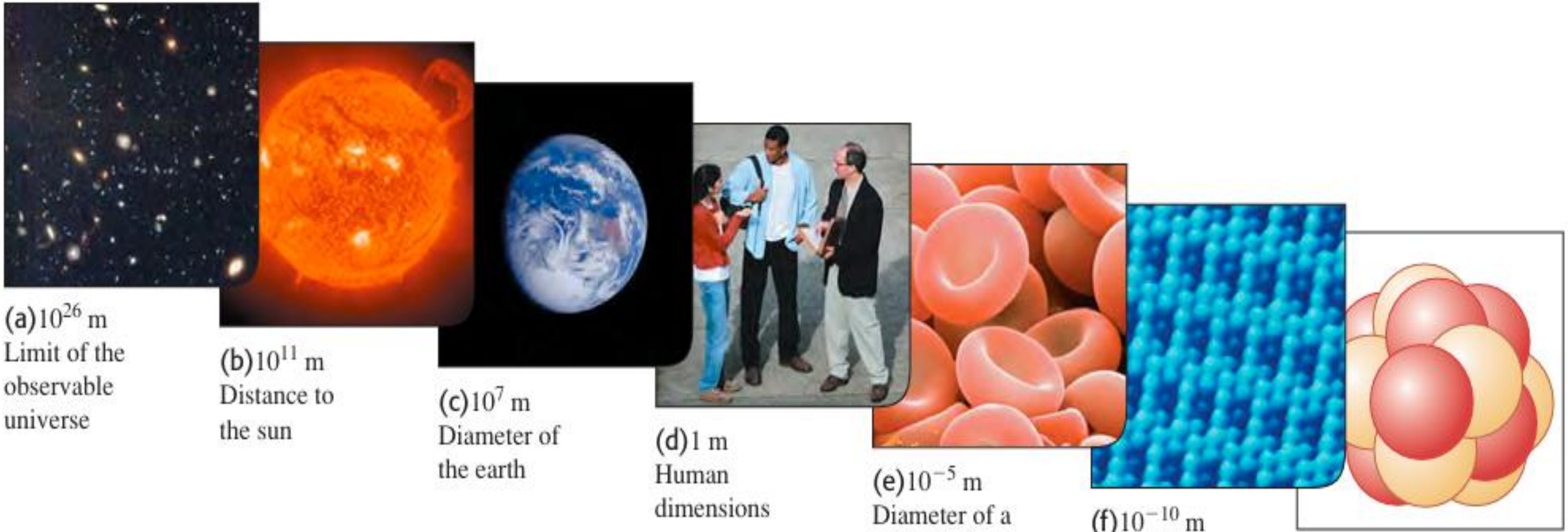
Submult	Prefix	Char	Mult	Prefix	Char
10^{-1}	deci	d	10	deca	da
10^{-2}	centi	c	10^2	hecto	h
10^{-3}	milli	m	10^3	kilo	k
10^{-6}	micro	u	10^6	mega	M
10^{-9}	nano	n	10^9	giga	G
10^{-12}	pico	p	10^{12}	tera	T
10^{-15}	femto	f	10^{15}	peta	P
10^{-18}	atto	a	10^{18}	exa	E
10^{-21}	zepto	z	10^{21}	zetta	Z
10^{-24}	yocto	y	10^{24}	yotta	Y

1 kilometer = 1 km = 10^3 meters = 10^3 m

1 kilogram = 1 kg = 10^3 grams = 10^3 g

1 kilowatt = 1 kW = 10^3 watts = 10^3 W

Some typical lengths in the universe



(a) 10^{26} m
Limit of the observable universe

(b) 10^{11} m
Distance to the sun

(c) 10^{7} m
Diameter of the earth

(d) 1 m
Human dimensions

(e) 10^{-5} m
Diameter of a red blood cell

(f) 10^{-10} m
Radius of an atom

(g) 10^{-14} m
Radius of an atomic nucleus

position

The position of an object tells us where the object is located in space at a particular time. To define a position clearly and unambiguously, we always need a reference.

The position of an object is its location relative to a chosen reference point, called the origin.

Definition:

Position is the location of an object with respect to a fixed reference point in a chosen coordinate system.

- Example:

Saying “The book is on the table” is not enough.

We must say, “The book is 50 cm to the right of the table’s center.”

The table’s center acts as the reference point (origin).

Coordinate System

To define position precisely, we use a coordinate system.

- (a) One-Dimensional (1D) Position

Used when motion is along a straight line (e.g., train on a track).

A straight line is chosen

One point is taken as the origin (O)

Position is given by a single number (x)

Example:

$x = +5$ m → object is 5 m to the right of the origin.

$X = -3$ m → object is 3 m to the left of the origin.

➢ Two-Dimensional (2D) Position

Used for motion in a plane (e.g., car on a road map).

Two coordinates give a position:

x-coordinate

y-coordinate

Example:

Position = (3 m, 4 m)

➢ Three-Dimensional (3D) Position

Used in space (e.g., flying bird, satellite).

Position is given by:

x, y, z coordinates

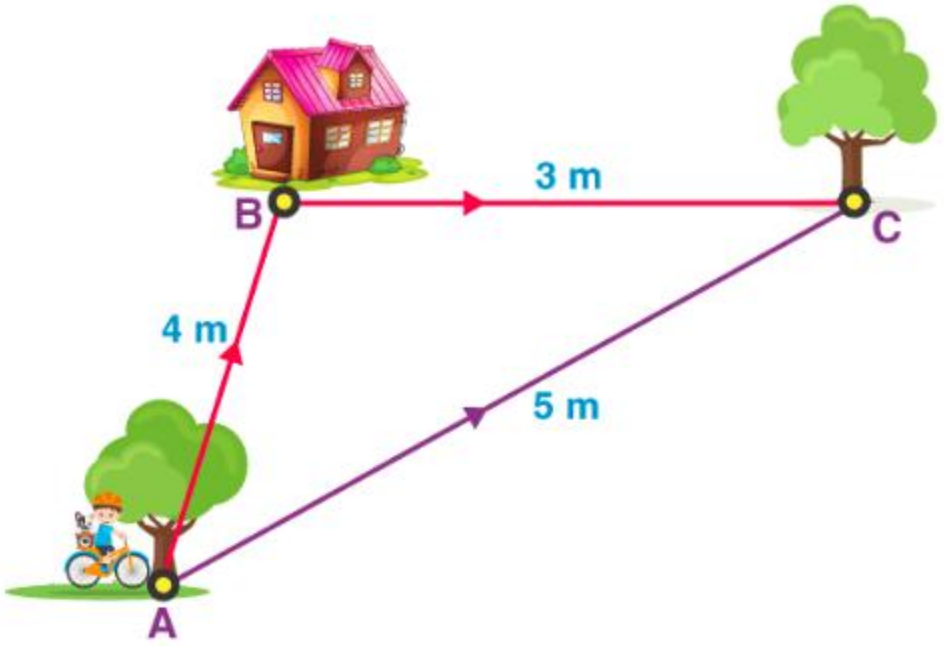
Example:

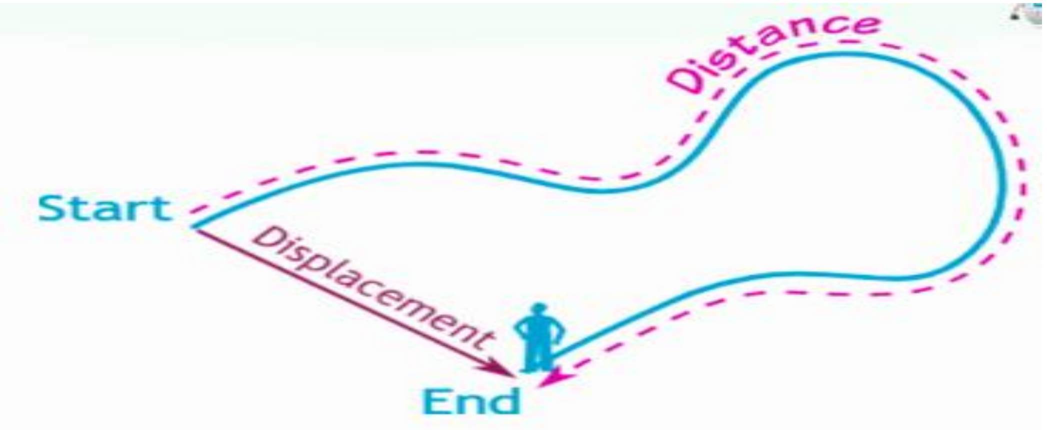
Position = (2 m, −1 m, 5 m)

Distance and Displacement

- Displacement refers to the change in position of an object or the distance and direction from its initial point to its final point.
- Distance involves the total ground covered, regardless of direction.

Distance and displacement are two quantities that seem to mean the same, but are distinctly different with different meanings and definitions. Distance is the measure of "how much ground an object has covered during its motion," while displacement refers to the measure of "how far out of place is an object."





Distance from A to C = 4m + 3m = 7 m
Displacement = 5m

Displacement is the change in position of an object in a particular direction.
Mathematically:
Displacement = Final Position − Initial Position

➢ Vector Nature of Displacement

Displacement is a vector quantity, which means:

It has a magnitude.
It has direction.

This makes displacement different from distance.

➢ Displacement Using Position Vectors

$$\vec{d} = \vec{r}_f - \vec{r}_i$$

Initial position vector = $\vec{r}_i$
Final position vector = $\vec{r}_f$

Average velocity

- Average velocity describes how fast and in which direction an object's position changes over a given time interval.
- Average velocity is defined as the total displacement divided by the total time taken.

Mathematically:

$$\text{Average Velocity} = \frac{\textit{Total displacement}}{\textit{Total time}} = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{t_f - t_i}$$

Initial position vector $= \vec{r}_i$
Final position vector $= \vec{r}_f$
Total time taken $= \Delta t$

$$\vec{v}_{avg} = \frac{\vec{r}_f - \vec{r}_i}{\Delta t}$$

Direction of Average Velocity

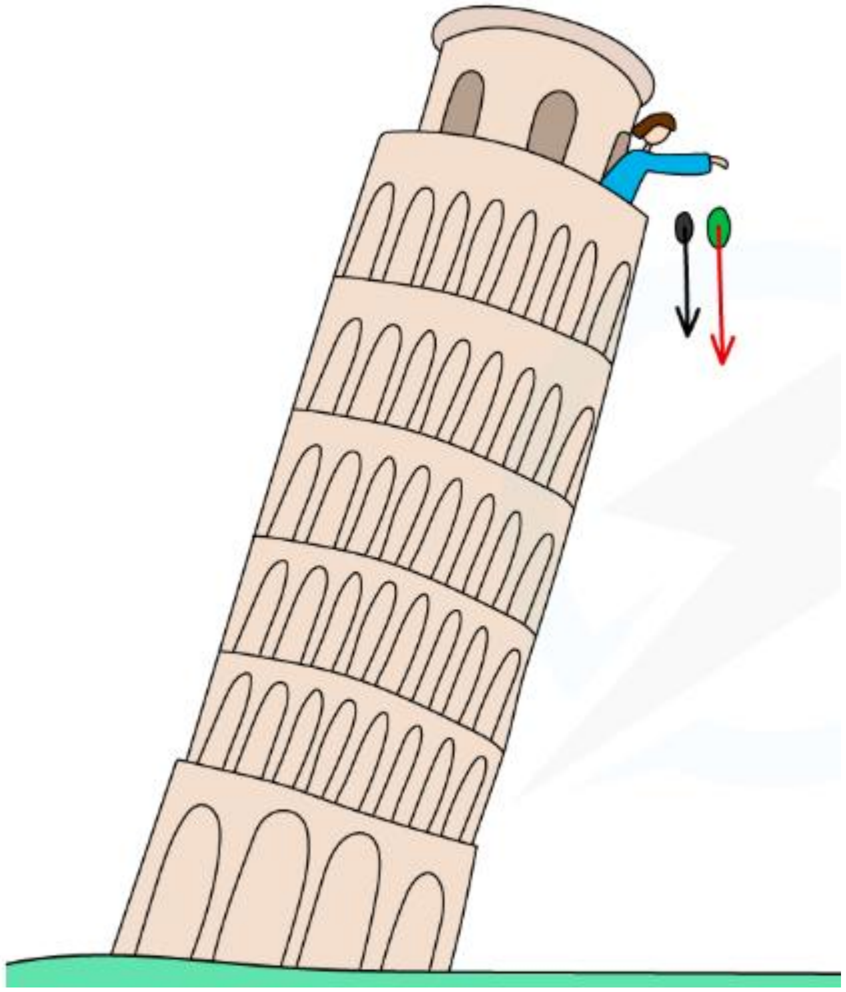
- The direction of average velocity is the same as the direction of displacement.
- If displacement is zero, average velocity is zero, even if the object was moving.

- CAUTION **The meaning of Δx**

- Note that Δx is not the product of Δ and x; it is a single symbol that means "the change in the quantity x." We always use the Greek capital letter (delta) to represent a change in a quantity, equal to the final value of the quantity minus the initial value — never the reverse. Likewise, the time interval from t1 to t2 change in the quantity t, $\Delta t = t_2 - t_1$(final time minus initial time).

Free-fall acceleration

- Free fall acceleration, or g, is the constant downward acceleration of objects due to gravity, approximately 9.8 m/s^2 (32 ft/s^2) at Earth's surface, meaning velocity increases by this amount each second.
- In a vacuum, all objects fall at this rate regardless of mass, but air resistance affects real-world objects, slowing them down and eventually leading to terminal velocity.

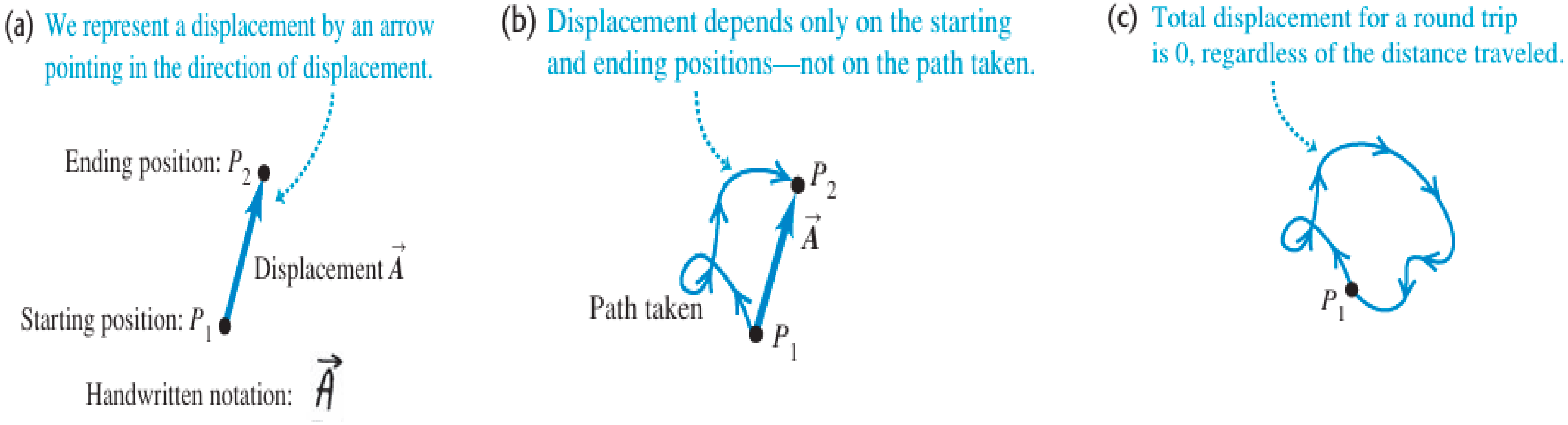


Normal Equations	Same equations specialized for freefall **a → g** and **x → y**
$v_f = v_i + at$ $v_f^2 = v_i^2 + 2a\Delta x$ $\Delta x = v_i t + 1/2\, at^2$	$v_{fy} = v_{iy} + gt$ $v_{fy}^2 = v_{iy}^2 + 2g\Delta y$ $\Delta y = v_{iy} t + 1/2\, gt^2$

Vectors

Scalar Quantity	Vector Quantity
1. Scalar quantity has only magnitude, but no direction.	1. Vector quantity has both magnitude and direction.
2. Every scalar quantity is one dimensional.	2. Vector quantity can be 1-D, 2-D or 3-D.
3. Any change in scalar quantity is the reflection of change in magnitude.	3. Any change in vector quantity can reflect either change in direction or change in magnitude or changes in both.
4. Scalar quantity cannot be resolved as it has exactly same value regardless of direction.	4. Vector can be resolved in any direction using sine or cosine of the adjacent angle.
5. Any mathematical operation carried out among two or more scalar quantities will provide a scalar only. However, if a scalar is operated with a vector then the result will be a vector.	5. Result of mathematical operations between two or more vectors may give either scalar or vector. Dot product of two vectors gives only scalar; while, cross product or summation or subtraction results in a vector.
6. Few examples of scalar quantity: length, mass, energy, density, etc.	6. Examples of vector quantity: displacement, velocity, acceleration, force, etc.

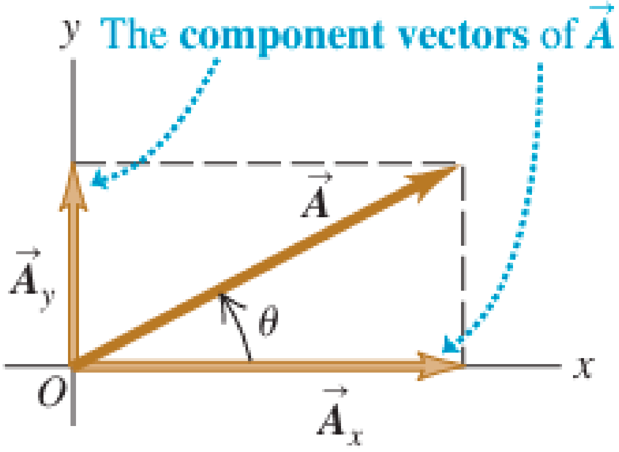
- Displacement as a vector quantity. A displacement is always a straight-line segment directed from the starting point to the ending point, even if the path is curved.

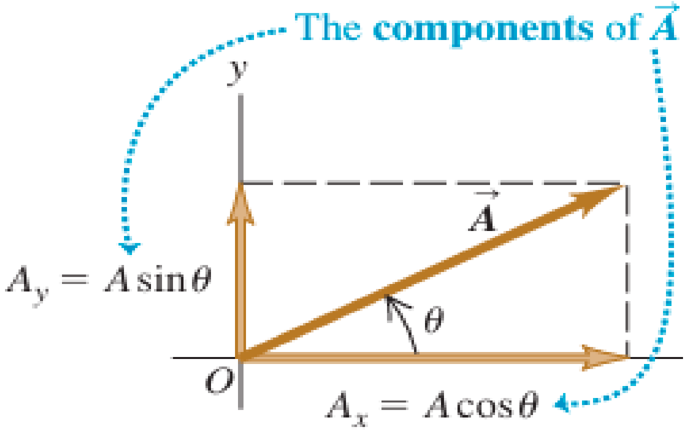


Components of Vectors

- Components of a vector are the parts of a vector along chosen coordinate axes (usually the x-axis and y-axis).
- A single vector can be replaced by two or three perpendicular vectors whose combined effect is the same as the original vector.

CAUTION: ***Components are not vectors*** The components A_x and A_y of a vector $\vec{A}$ are just numbers; they are not vectors themselves. This is why we print the symbols for components in light italic type with no arrow on top instead of in boldface italic with an arrow, which is reserved for vectors.





$$\vec{A} = \vec{A}_x + \vec{A}_y$$

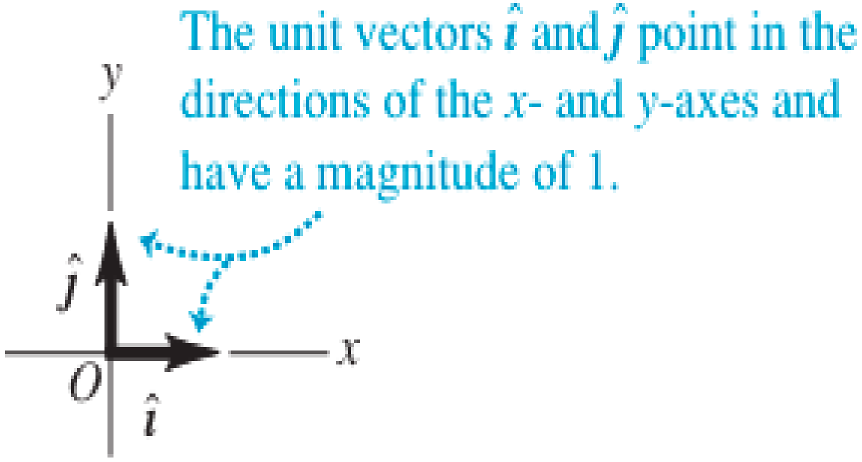
$$\frac{A_x}{A} = \cos\theta \quad \text{and} \quad \frac{A_y}{A} = \sin\theta$$

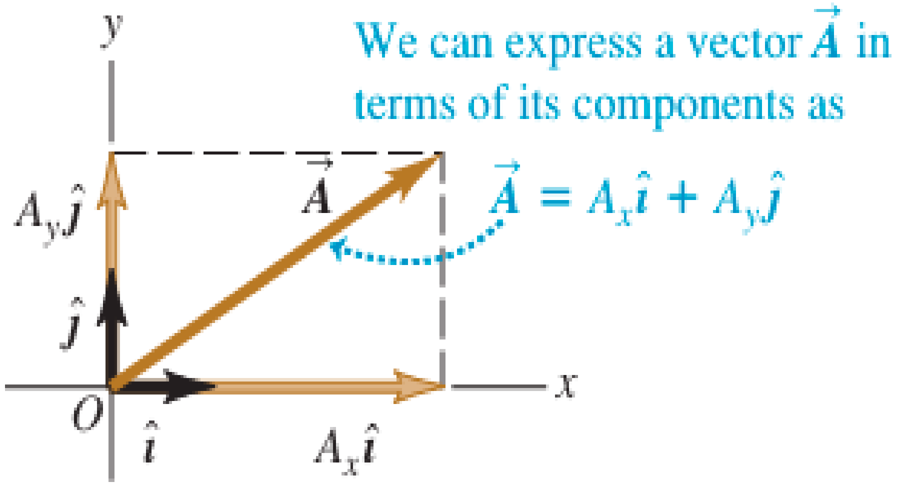
$$A_x = A\cos\theta \quad \text{and} \quad A_y = A\sin\theta$$

(θ measured from the $+x$-axis, rotating toward the $+y$-axis)

unit vector

- A unit vector is a vector that has a magnitude of 1, with no units.
- Its only purpose is to point—that is, to describe a direction in space.
- Unit vectors provide a convenient notation for many expressions involving components of vectors.
- We will always include a caret or “hat” in the symbol for a unit vector to distinguish it from ordinary vectors whose magnitude may or may not be equal to 1.





$$\vec{A}_x = A_x\hat{\imath}$$
$$\vec{A}_y = A_y\hat{\jmath}$$

$$\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath}$$

$$\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath}$$

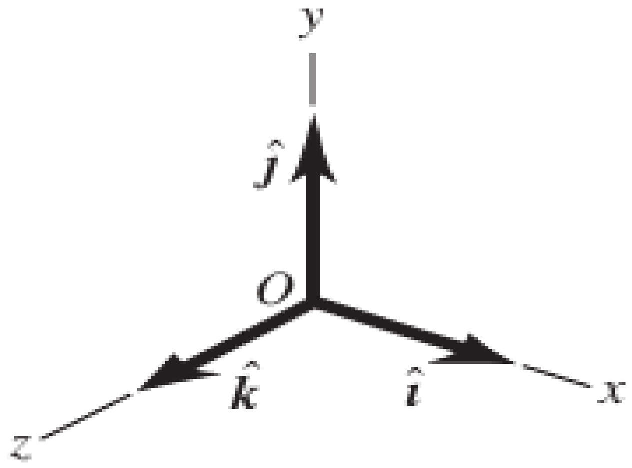
$$\vec{B} = B_x\hat{\imath} + B_y\hat{\jmath}$$

$$\begin{aligned}\vec{R} &= \vec{A} + \vec{B} \\ &= (A_x\hat{\imath} + A_y\hat{\jmath}) + (B_x\hat{\imath} + B_y\hat{\jmath}) \\ &= (A_x + B_x)\hat{\imath} + (A_y + B_y)\hat{\jmath} \\ &= R_x\hat{\imath} + R_y\hat{\jmath}\end{aligned}$$

$$\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k}$$

$$\vec{B} = B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k}$$

$$\begin{aligned}\vec{R} &= (A_x + B_x)\hat{\imath} + (A_y + B_y)\hat{\jmath} + (A_z + B_z)\hat{k} \\ &= R_x\hat{\imath} + R_y\hat{\jmath} + R_z\hat{k}\end{aligned}$$



Example

Given the two displacements

$$\vec{D} = (6.00\hat{\imath} + 3.00\hat{\jmath} - 1.00\hat{k})\ \text{m} \quad \text{and}$$

$$\vec{E} = (4.00\hat{\imath} - 5.00\hat{\jmath} + 8.00\hat{k})\ \text{m}$$

find the magnitude of the displacement $2\vec{D} - \vec{E}$.

$$\vec{F} = 2(6.00\hat{\imath} + 3.00\hat{\jmath} - 1.00\hat{k})\ \text{m} - (4.00\hat{\imath} - 5.00\hat{\jmath} + 8.00\hat{k})\ \text{m}$$

$$= [(12.00 - 4.00)\hat{\imath} + (6.00 + 5.00)\hat{\jmath} + (-2.00 - 8.00)\hat{k}]\ \text{m}$$

$$= (8.00\hat{\imath} + 11.00\hat{\jmath} - 10.00\hat{k})\ \text{m}$$

From Eq. (1.12) the magnitude of $\vec{F}$ is

$$F = \sqrt{F_x{}^2 + F_y{}^2 + F_z{}^2}$$

$$= \sqrt{(8.00\ \text{m})^2 + (11.00\ \text{m})^2 + (-10.00\ \text{m})^2}$$

$$= 16.9\ \text{m}$$

Arrange the following vectors in order of their magnitude, with the vector of largest magnitude first.
(i) A = (3i+5j - 2k) m
(ii) B = (-3i + 5j - 2k) m
(iii) C = (3i - 5j -2k) m
(iv) D = (3i+ 5J + 2k) m.

To arrange the vectors by magnitude, we calculate the magnitude of each vector using:

$$|\vec{V}| = \sqrt{x^2 + y^2 + z^2}$$

$$\vec{A} = (3i + 5j - 2k)$$

$$|\vec{A}| = \sqrt{3^2 + 5^2 + (-2)^2} = \sqrt{9 + 25 + 4} = \sqrt{38}$$

$$\vec{B} = (-3i + 5j - 2k)$$

$$|\vec{B}| = \sqrt{(-3)^2 + 5^2 + (-2)^2} = \sqrt{38}$$

$$\vec{C} = (3i - 5j - 2k)$$

$$|\vec{C}| = \sqrt{3^2 + (-5)^2 + (-2)^2} = \sqrt{38}$$

$$\vec{D} = (3i + 5j + 2k)$$

$$|\vec{D}| = \sqrt{3^2 + 5^2 + 2^2} = \sqrt{38}$$

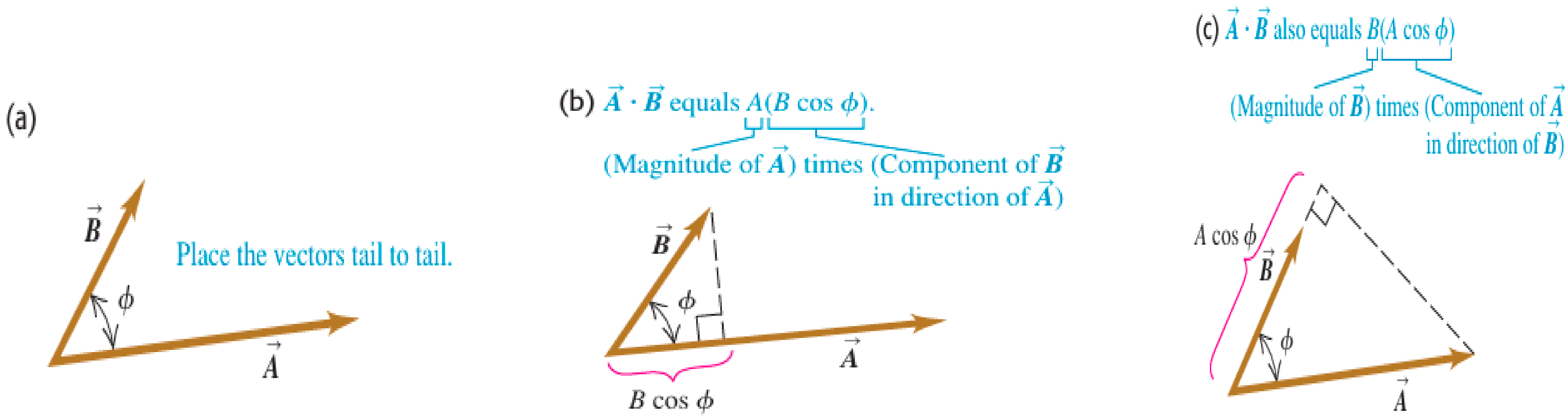
$$\boxed{A = B = C = D}$$

All four vectors have **equal magnitude**.

Product of vectors

Scalar Product

The scalar product of two vectors $\vec{\boldsymbol{A}}$ and $\vec{\boldsymbol{B}}$ is denoted By $\vec{\boldsymbol{A}}.\vec{\boldsymbol{B}}$. Because of this notation, the scalar product is also called the dot product. Although $\vec{\boldsymbol{A}}$ and $\vec{\boldsymbol{B}}$ are vectors, the quantity $\vec{\boldsymbol{A}}.\vec{\boldsymbol{B}}$ is a scalar.



- To define the scalar product $\vec{A}.\vec{B}$ we draw the two vectors $\vec{A}$ and $\vec{B}$ with their tails at the same point. The angle φ (the Greek letter phi) between their directions ranges from 0°to 180°. The projection of the vector onto the direction of this projection is the component of in the direction of and is equal to (We can take components along any direction that's convenient, not just the x- and y-axes.) We define to be the magnitude of multiplied by the component of $\vec{B}$ in the direction of $\vec{A}$.

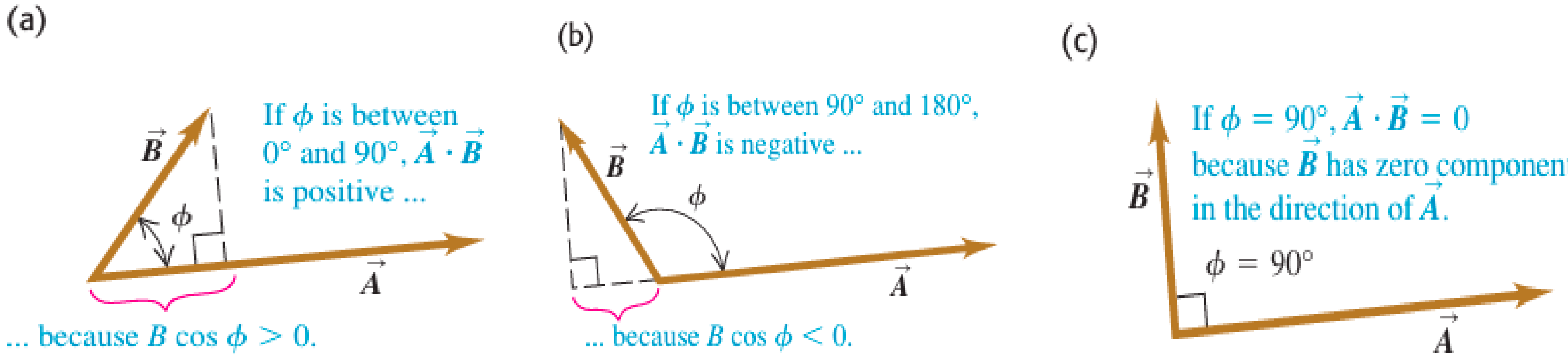
$$\vec{A} \cdot \vec{B} = AB\cos\phi = |\vec{A}||\vec{B}|\cos\phi \quad \text{(definition of the scalar (dot) product)}$$

The scalar product is a scalar quantity, not a vector, and it may be positive, negative, or zero. When φ is between 0°and 90^0, cos φ > 0 and the scalar product is positive.

When φ is between 90° and 180^0 so that cos φ < 0, the component of $\vec{B}$ in the direction of $\vec{A}$ is negative, and $\vec{A}.\ \vec{B}$ *is* negative.

Finally, when φ = 90^0, $\vec{A}.\ \vec{B}=0$ The scalar product of two perpendicular vectors is always zero.

For any two vectors $\vec{A}$ and $\vec{B}$, AB cos φ = BA cos φ. This means that $\vec{A}.\ \vec{B}=\vec{B}.\ \vec{A}$ The scalar product obeys the commutative law of multiplication; the order of the two vectors does not matter.



Calculating the Scalar Product Using Components

$$
\begin{aligned}
\vec{A}\cdot\vec{B} &= (A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k})\cdot(B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k})\\
&= A_x\hat{\imath}\cdot B_x\hat{\imath} + A_x\hat{\imath}\cdot B_y\hat{\jmath} + A_x\hat{\imath}\cdot B_z\hat{k}\\
&\quad + A_y\hat{\jmath}\cdot B_x\hat{\imath} + A_y\hat{\jmath}\cdot B_y\hat{\jmath} + A_y\hat{\jmath}\cdot B_z\hat{k}\\
&\quad + A_z\hat{k}\cdot B_x\hat{\imath} + A_z\hat{k}\cdot B_y\hat{\jmath} + A_z\hat{k}\cdot B_z\hat{k}\\
&= A_xB_x\hat{\imath}\cdot\hat{\imath} + A_xB_y\hat{\imath}\cdot\hat{\jmath} + A_xB_z\hat{\imath}\cdot\hat{k}\\
&\quad + A_yB_x\hat{\jmath}\cdot\hat{\imath} + A_yB_y\hat{\jmath}\cdot\hat{\jmath} + A_yB_z\hat{\jmath}\cdot\hat{k}\\
&\quad + A_zB_x\hat{k}\cdot\hat{\imath} + A_zB_y\hat{k}\cdot\hat{\jmath} + A_zB_z\hat{k}\cdot\hat{k}
\end{aligned}
$$

$$\hat{\imath}\cdot\hat{\imath} = \hat{\jmath}\cdot\hat{\jmath} = \hat{k}\cdot\hat{k} = (1)(1)\cos 0^\circ = 1$$

$$\hat{\imath}\cdot\hat{\jmath} = \hat{\imath}\cdot\hat{k} = \hat{\jmath}\cdot\hat{k} = (1)(1)\cos 90^\circ = 0$$

Thus, the scalar product of two vectors is the sum of the products of their respective components.

$$\vec{A}\cdot\vec{B} = A_xB_x + A_yB_y + A_zB_z$$ (scalar (dot) product in terms of components)

Finding an angle with the scalar product

Find the angle between the vectors

$$\vec{A} = 2.00\hat{\imath} + 3.00\hat{\jmath} + 1.00\hat{k} \qquad \text{and}$$
$$\vec{B} = -4.00\hat{\imath} + 2.00\hat{\jmath} - 1.00\hat{k}$$

$$\cos\phi = \frac{\vec{A}\cdot\vec{B}}{AB} = \frac{A_xB_x + A_yB_y + A_zB_z}{AB}$$

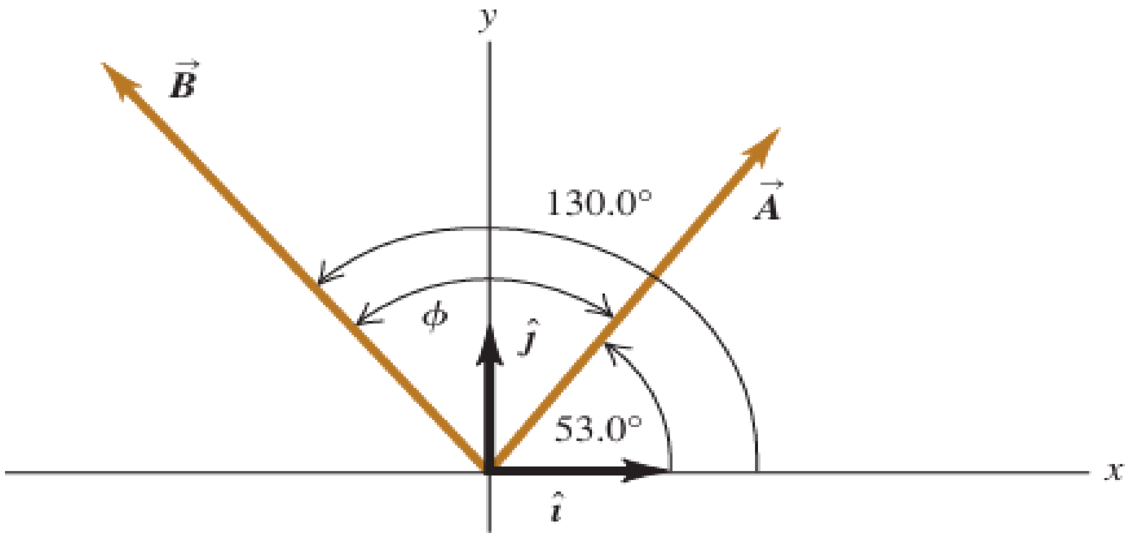
$$\begin{aligned}
\vec{A}\cdot\vec{B} &= A_xB_x + A_yB_y + A_zB_z \\
&= (2.00)(-4.00) + (3.00)(2.00) + (1.00)(-1.00) \\
&= -3.00 \\
A &= \sqrt{A_x^2 + A_y^2 + A_z^2} = \sqrt{(2.00)^2 + (3.00)^2 + (1.00)^2} \\
&= \sqrt{14.00} \\
B &= \sqrt{B_x^2 + B_y^2 + B_z^2} = \sqrt{(-4.00)^2 + (2.00)^2 + (-1.00)^2} \\
&= \sqrt{21.00} \\
\cos\phi &= \frac{A_xB_x + A_yB_y + A_zB_z}{AB} = \frac{-3.00}{\sqrt{14.00}\,\sqrt{21.00}} = -0.175 \\
\phi &= 100^\circ
\end{aligned}$$

Example 1.10 Calculating a scalar product

Find the scalar product $\vec{A} \cdot \vec{B}$ of the two vectors in Fig. 1.27. The magnitudes of the vectors are $A = 4.00$ and $B = 5.00$.

SOLUTION

IDENTIFY and SET UP: We can calculate the scalar product in two ways: using the magnitudes of the vectors and the angle between them (Eq. 1.18), and using the components of the vectors (Eq. 1.21). We'll do it both ways, and the results will check each other.



EXECUTE: The angle between the two vectors is $\phi = 130.0° - 53.0° = 77.0°$, so Eq. (1.18) gives us

$$\vec{A} \cdot \vec{B} = AB\cos\phi = (4.00)(5.00)\cos 77.0° = 4.50$$

To use Eq. (1.21), we must first find the components of the vectors. The angles of $\vec{A}$ and $\vec{B}$ are given with respect to the $+x$-axis and are measured in the sense from the $+x$-axis to the $+y$-axis, so we can use Eqs. (1.6):

$$A_x = (4.00)\cos 53.0° = 2.407$$
$$A_y = (4.00)\sin 53.0° = 3.195$$
$$B_x = (5.00)\cos 130.0° = -3.214$$
$$B_y = (5.00)\sin 130.0° = 3.830$$

As in Example 1.7, we keep an extra significant figure in the components and round at the end. Equation (1.21) now gives us

$$\begin{aligned}\vec{A} \cdot \vec{B} &= A_xB_x + A_yB_y + A_zB_z \\ &= (2.407)(-3.214) + (3.195)(3.830) + (0)(0) = 4.50\end{aligned}$$

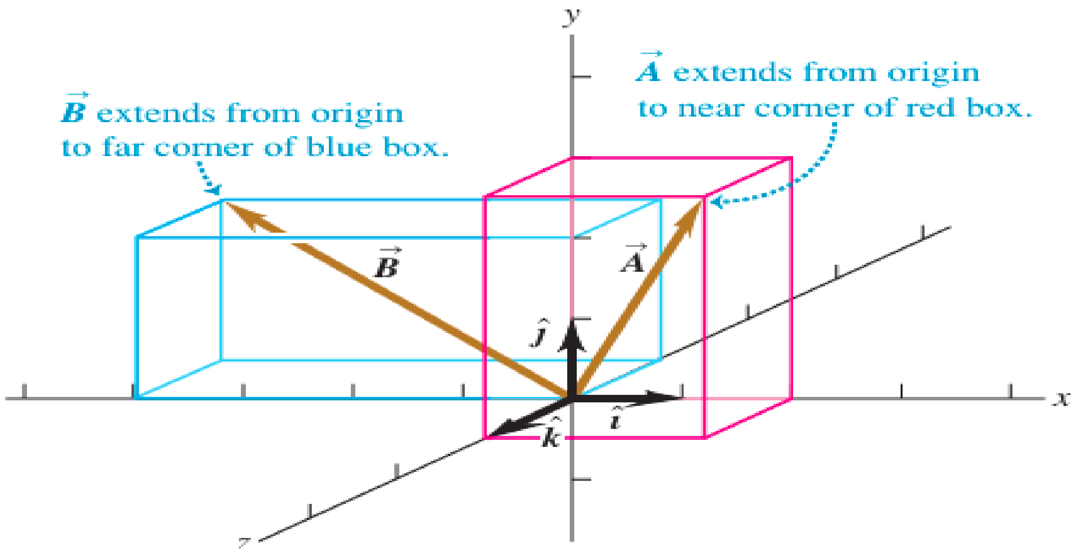
Example 1.11 Finding an angle with the scalar product

Find the angle between the vectors

$$\vec{A} = 2.00\hat{\imath} + 3.00\hat{\jmath} + 1.00\hat{k} \qquad \text{and}$$
$$\vec{B} = -4.00\hat{\imath} + 2.00\hat{\jmath} - 1.00\hat{k}$$

SOLUTION

IDENTIFY and SET UP: We're given the x-, y-, and z-components of two vectors. Our target variable is the angle ϕ between them (Fig. 1.28). To find this, we'll solve Eq. (1.18), $\vec{A}\cdot\vec{B} = AB\cos\phi$, for ϕ in terms of the scalar product $\vec{A}\cdot\vec{B}$ and the magnitudes A and B. We can evaluate the scalar product using Eq. (1.21), $\vec{A}\cdot\vec{B} = A_xB_x + A_yB_y + A_zB_z$, and we can find A and B using Eq. (1.7).



EXECUTE: We solve Eq. (1.18) for $\cos\phi$ and write $\vec{A}\cdot\vec{B}$ using Eq. (1. 21). Our result is

$$\cos\phi = \frac{\vec{A}\cdot\vec{B}}{AB} = \frac{A_xB_x + A_yB_y + A_zB_z}{AB}$$

We can use this formula to find the angle between *any* two vectors $\vec{A}$ and $\vec{B}$. Here we have $A_x = 2.00$, $A_y = 3.00$, and $A_z = 1.00$, and $B_x = -4.00$, $B_y = 2.00$, and $B_z = -1.00$. Thus

$$\begin{aligned}
\vec{A}\cdot\vec{B} &= A_xB_x + A_yB_y + A_zB_z \\
&= (2.00)(-4.00) + (3.00)(2.00) + (1.00)(-1.00) \\
&= -3.00 \\
A &= \sqrt{A_x^2 + A_y^2 + A_z^2} = \sqrt{(2.00)^2 + (3.00)^2 + (1.00)^2} \\
&= \sqrt{14.00} \\
B &= \sqrt{B_x^2 + B_y^2 + B_z^2} = \sqrt{(-4.00)^2 + (2.00)^2 + (-1.00)^2} \\
&= \sqrt{21.00} \\
\cos\phi &= \frac{A_xB_x + A_yB_y + A_zB_z}{AB} = \frac{-3.00}{\sqrt{14.00}\,\sqrt{21.00}} = -0.175 \\
\phi &= 100^\circ
\end{aligned}$$

Vector Product

- The **vector product** (or **cross product**) of two vectors $\vec{A}$ and $\vec{B}$ is written as $\vec{A} \times \vec{B}$.
- The result of a cross product is **also a vector.**
- It is used to describe physical quantities like **torque, angular momentum, magnetic fields, and magnetic forces.**
- To define the cross product:
    - Place $\vec{A}$ and $\vec{B}$ tail-to-tail; they lie in a plane.
    - The direction of $\vec{A} \times \vec{B}$ is **perpendicular to the plane** containing $\vec{A}$ and $\vec{B}$.
- The **magnitude** of the cross product is:

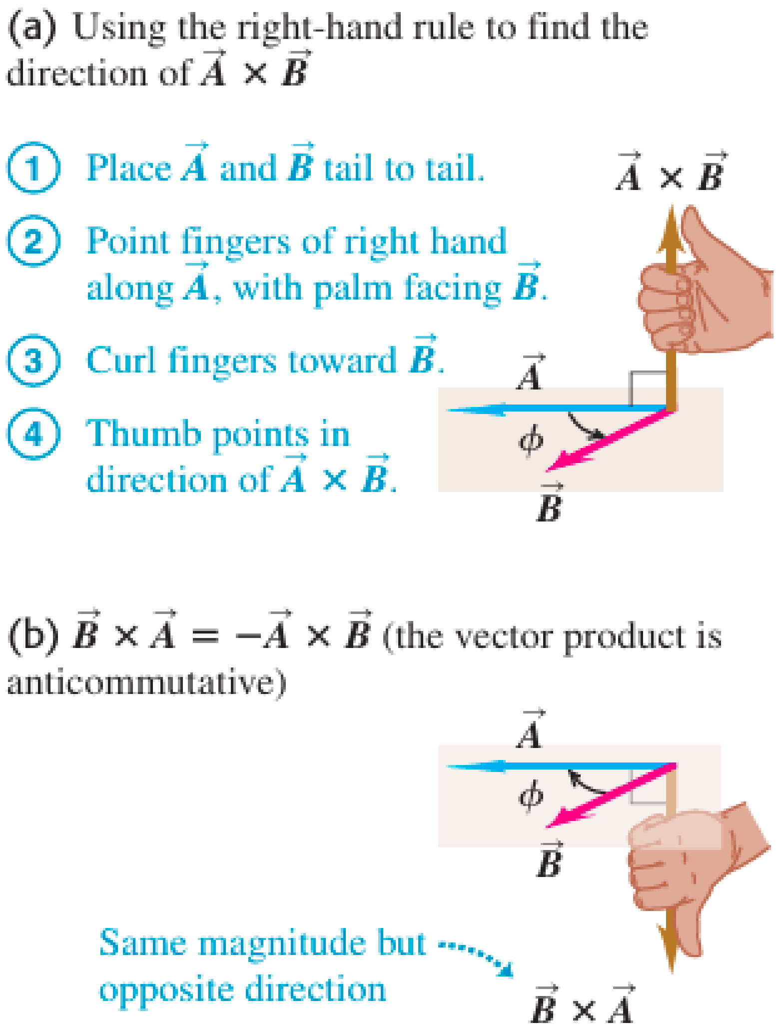
$$|\vec{A} \times \vec{B}| = AB \sin \phi$$

where ϕ is the angle between $\vec{A}$ and $\vec{B}$.

- The angle ϕ is measured from $\vec{A}$ **toward** $\vec{B}$ and lies between **0° and 180°**.
- Since $\sin\phi \geq 0$, the magnitude of the cross product is **never negative**.
- If $\vec{A}$ and $\vec{B}$ are **parallel or antiparallel** ($\phi = 0^\circ$ or 180°):
  - $\sin\phi = 0$
  - The cross product is **zero**.
- The cross product of **any vector with itself is zero**.

- For any plane, there are **two possible perpendicular directions** (one on each side).
- The direction of $\vec{A} \times \vec{B}$ is chosen using the **right-hand rule**.
- **Right-hand rule**:
  - Rotate vector $\vec{A}$ toward $\vec{B}$ through the **smaller angle**.
  - Curl the fingers of your **right hand** in the direction of rotation.
  - The **thumb points in the direction of** $\vec{A} \times \vec{B}$.
- To find the direction of $\vec{B} \times \vec{A}$:
  - Rotate $\vec{B}$ toward $\vec{A}$.
  - The resulting direction is **opposite** to that of $\vec{A} \times \vec{B}$.
- Therefore, the cross product is **not commutative**:

1.29 (a) The vector product $\vec{A} \times \vec{B}$ determined by the right-hand rule.
(b) $\vec{B} \times \vec{A} = -\vec{A} \times \vec{B}$; the vector product is anticommutative.



$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$$

- **Geometrical meaning of magnitude:**
  - $B \sin \phi$ is the component of $\vec{B}$ **perpendicular** to $\vec{A}$.
  - The magnitude of the cross product is:

$$|\vec{A} \times \vec{B}| = A \times (B \sin \phi)$$

- Similarly:
  - The magnitude also equals B times the component of $\vec{A}$ perpendicular to $\vec{B}$.
- This geometric interpretation is valid for **all angles between 0° and 180°**.

Unit vectors product

$$\hat{\imath} \times \hat{\imath} = \hat{\jmath} \times \hat{\jmath} = \hat{k} \times \hat{k} = 0$$

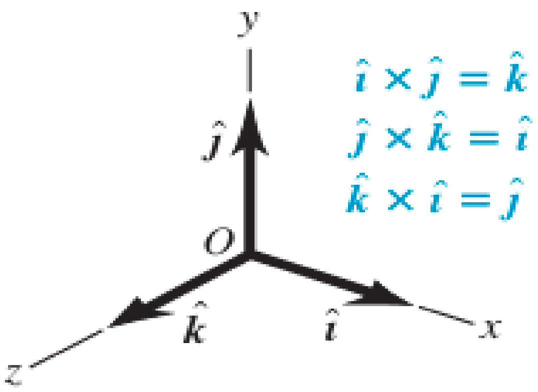
$$\hat{\imath} \times \hat{\jmath} = -\hat{\jmath} \times \hat{\imath} = \hat{k}$$

$$\hat{\jmath} \times \hat{k} = -\hat{k} \times \hat{\jmath} = \hat{\imath}$$

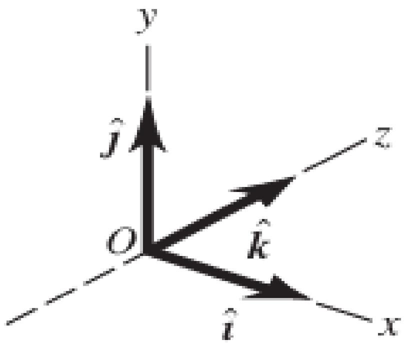
$$\hat{k} \times \hat{\imath} = -\hat{\imath} \times \hat{k} = \hat{\jmath}$$

$$\begin{aligned}\vec{A} \times \vec{B} &= (A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k}) \times (B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k}) \\ &= A_x\hat{\imath} \times B_x\hat{\imath} + A_x\hat{\imath} \times B_y\hat{\jmath} + A_x\hat{\imath} \times B_z\hat{k} \\ &\quad + A_y\hat{\jmath} \times B_x\hat{\imath} + A_y\hat{\jmath} \times B_y\hat{\jmath} + A_y\hat{\jmath} \times B_z\hat{k} \\ &\quad + A_z\hat{k} \times B_x\hat{\imath} + A_z\hat{k} \times B_y\hat{\jmath} + A_z\hat{k} \times B_z\hat{k}\end{aligned}$$

(a) A right-handed coordinate system



(b) A left-handed coordinate system; we will not use these.



$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y)\hat{\imath} + (A_z B_x - A_x B_z)\hat{\jmath} + (A_x B_y - A_y B_x)\hat{k}$$

Thus the components of $\vec{C} = \vec{A} \times \vec{B}$ are given by

$$C_x = A_y B_z - A_z B_y \qquad C_y = A_z B_x - A_x B_z \qquad C_z = A_x B_y - A_y B_x$$

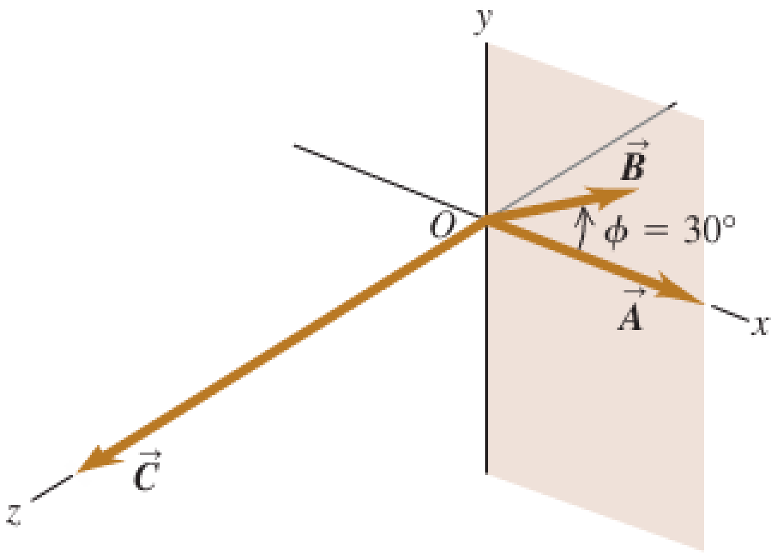
(components of $\vec{C} = \vec{A} \times \vec{B}$)

The vector product can also be expressed in determinant form as

$$\vec{A} \times \vec{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

Example 1.12 Calculating a vector product

Vector $\vec{A}$ has magnitude 6 units and is in the direction of the $+x$-axis. Vector $\vec{B}$ has magnitude 4 units and lies in the xy-plane, making an angle of 30° with the $+x$-axis (Fig. 1.32). Find the vector product $\vec{C} = \vec{A} \times \vec{B}$.



$$AB \sin \phi = (6)(4)(\sin 30°) = 12$$

By the right-hand rule, the direction of $\vec{A} \times \vec{B}$ is along the $+z$-axis (the direction of the unit vector $\hat{k}$), so we have $\vec{C} = \vec{A} \times \vec{B} = 12\hat{k}$.

To use Eqs. (1.27), we first determine the components of $\vec{A}$ and $\vec{B}$:

$$A_x = 6 \qquad A_y = 0 \qquad A_z = 0$$
$$B_x = 4 \cos 30° = 2\sqrt{3} \qquad B_y = 4 \sin 30° = 2 \qquad B_z = 0$$

Then Eqs. (1.27) yield

$$C_x = (0)(0) - (0)(2) = 0$$
$$C_y = (0)(2\sqrt{3}) - (6)(0) = 0$$
$$C_z = (6)(2) - (0)(2\sqrt{3}) = 12$$

Thus again we have $\vec{C} = 12\hat{k}$.

EVALUATE: Both methods give the same result. Depending on the situation, one or the other of the two approaches may be the more convenient one to use.

Test Your Understanding of Vector Multiplications

Vector $\vec{\boldsymbol{A}}$ has magnitude 2 and vector $\vec{\boldsymbol{B}}$ has magnitude 3. The angle φ between $\vec{\boldsymbol{A}}$ and $\vec{\boldsymbol{B}}$ is known to be 0°, 90°, or 180°. For each of the following situations, state what the value of φ must be. (In each situation, there may be more than one correct answer.) (a) $\vec{\boldsymbol{A}}.\vec{\boldsymbol{B}} = 0$ (b) $\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}} = 0$ (c) $\vec{\boldsymbol{A}}.\vec{\boldsymbol{B}} = 6$

(d) $\vec{\boldsymbol{A}}.\vec{\boldsymbol{B}} = -6$ (e) (Magnitude of $\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}} = 6$

1. Can you find two vectors with different lengths that have a vector sum of zero? What length restrictions are required for three vectors to have a vector sum of zero? Explain your reasoning.
2. One sometimes speaks of the “direction of time,” evolving from past to future. Does this mean that time is a vector quantity? Explain your reasoning.
3. Air traffic controllers give instructions to airline pilots, telling them in which direction they are to fly. These instructions are called “vectors.” If these are the only instructions given, in the name “vector” used correctly? Why or why not?
4. Can you find a vector quantity that has a magnitude of zero but components that are different from zero?
5. Explain. Can the magnitude of a vector be less than the magnitude of any of its components? Explain.
6. Does it make sense to say that a vector is negative? Why? (b) Does it make sense to say that one vector is the negative of another? Why? Does your answer here contradict what you said in part (a)?

Projectile Motion

Projectile motion is the motion of an object that is thrown or projected into the air and then moves only under the influence of gravity (air resistance neglected).

Examples:

- A ball thrown at an angle
- A stone thrown horizontally from a building
- A bullet fired from a gun
- A football kicked into the air
- Once the object is projected, no force acts on it except gravity.

Basic Assumptions in Projectile Motion

- To simplify analysis, we assume:
- Air resistance is negligible
- Acceleration due to gravity, g is constant
- The Earth is flat over the range of motion
- Motion takes place near the surface of the Earth.

Analysis of Projectile Motion (Key Idea)

- Projectile motion can be divided into two independent motions:
  - Horizontal motion
  - Vertical motion

These two motions occur simultaneously but are independent of each other, connected only by time.

Horizontal Motion of a Projectile

There is no acceleration in the horizontal direction (if air resistance is neglected).

Horizontal velocity remains constant throughout the motion.

Horizontal velocity:

If the projectile is thrown with initial velocity $u_x = u\cos\theta$

Horizontal displacement:
$X = u\cos\theta t$

Horizontal motion is uniform motion
Follows the laws of motion with constant velocity.

Vertical Motion of a Projectile

- Vertical motion is affected by gravity.
- Acceleration is always downward and equal to g.

Initial vertical velocity:

$$uy = u\sin\theta$$

Vertical velocity at time t. $v_y = u\sin\theta - gt$

Vertical displacement: $y = u\sin\theta t - 1/2gt^2$

So:

- Vertical motion is uniformly accelerated motion.
- Same as motion under gravity (free fall).